ModelIt!

Student Activity Pack

Can We Edit the Human Genome?

Modeling CRISPR Gene Editing, Inheritance, and the Ethics of Genetic Engineering

Grade 12L1 | NGSS HS-LS3-1 | Structure and Function

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you already know about CRISPR or gene editing? Where did you learn about it?

2 If scientists could edit a gene to prevent a fatal disease in an unborn child, do you think they should? Why or why not?

3 What's the difference between curing a disease and enhancing a trait? Is the line always clear?

4 Who should decide what gene edits are allowed — scientists, governments, parents, or society as a whole?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Guide RNA Specificity | Target Gene Location | Editing Efficiency | Off-Target Mutation Rate | Therapeutic Outcome

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Sickle Cell Cure

Guide RNA Specificity: High | Target: Single gene (HBB) | Delivery: Somatic cells

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Multiple Gene Edit

Guide RNA Specificity: Moderate | Target: 5 genes simultaneously | Delivery: Somatic cells

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Germline Enhancement

Guide RNA Specificity: High | Target: Multiple trait genes | Delivery: Embryo (germline)

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Delivery Method Efficiency — How effectively the CRISPR components are delivered into target cells — viral vectors, lipid nanoparticles, and electroporation each have different efficiency rates and tissue specificity that affect overall therapeutic outcome

Mosaicism Risk — The probability that only some cells in an organism are successfully edited while others retain the original gene — creates organisms with mixed genetic populations, reducing therapeutic effectiveness

Epigenetic Disruption — The risk that DNA cuts or insertions alter the epigenetic marks (methylation, histone modification) near the target site, changing gene expression patterns in ways that may not be apparent for years

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

CRISPR technology lets scientists edit DNA like a word processor edits text — cutting out disease-causing genes and replacing them with healthy versions. If we can cure genetic diseases before birth, should we? And where is the line between healing and designing 'perfect' humans?
Use evidence from your simulation to support your answers.

1. **How does your model help explain why CRISPR works well for single-gene diseases but struggles with complex traits like intelligence or height?**

2. **What does the Off-Target Mutation Rate reveal about why we should be cautious even when editing seems highly precise?**

3. **Should germline editing be banned, regulated, or left to individual families? What evidence from your model informs your position?**

4. **If gene editing becomes affordable, how might it affect social inequality? What if only wealthy families can access genetic enhancements?**

5. **How does the precautionary principle apply to a technology that could cure devastating diseases but also enable genetic discrimination?**

8 STEM Challenge: Design a Gene Therapy Ethics Framework

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YOUR ENGINEERING MISSION

Design a comprehensive ethical framework that a hospital genetics board could use to evaluate whether a proposed CRISPR gene-editing treatment should be approved, considering both scientific evidence and societal implications.

THE SCENARIO

A genetics company has developed a CRISPR treatment that could eliminate the gene for Huntington's disease in embryos — a fatal neurodegenerative disorder with no cure. But the same technology could potentially be used to select for traits like height, intelligence, or eye color. Your hospital's ethics board needs a decision framework that distinguishes between therapeutic and enhancement uses of gene editing.

Engineering Design Thinking

1. WHAT SPECIFIC CRITERIA DOES YOUR FRAMEWORK USE TO EVALUATE WHETHER AN EDIT IS 'THERAPEUTIC' VS. 'ENHANCEMENT'?

2. HOW DOES YOUR FRAMEWORK ACCOUNT FOR THE SCIENTIFIC UNCERTAINTY OF OFF-TARGET EFFECTS WHEN MAKING APPROVAL DECISIONS?

3. WHAT SAFEGUARDS DOES YOUR FRAMEWORK INCLUDE FOR GERMLINE EDITS THAT WILL AFFECT FUTURE GENERATIONS?

4. HOW DOES YOUR FRAMEWORK ADDRESS EQUITY — PREVENTING A FUTURE WHERE GENE EDITING CREATES BIOLOGICAL INEQUALITY BETWEEN RICH AND POOR?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Genetic counselors help families understand genetic testing results and make informed decisions about treatments including gene therapy, earning $80,000-$110,000/year. Bioethicists specializing in genetic technology work at hospitals, universities, and policy organizations, earning $60,000-$130,000/year. CRISPR researchers in biotechnology companies earn $90,000-$180,000/year.